POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 2031 | Semester II | Theory | 4 Credits

Elementary Concepts of Electrical & Electronics Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering, Electrical Engineering and Electrical Engineering & Electric Vehicles Technology**
Contact hours	**60**
Teaching scheme	**4:0:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**6 h/week; 90 h total**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2031**, titled **Elementary Concepts of Electrical & Electronics Engineering**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

Source treatment: PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

Verified source note: No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

This course introduces electrical energy use and conservation, energy-conversion devices, magnetism and magnetic circuits, and bipolar-transistor operation. It connects domestic and industrial applications to calculation and safe engineering practice.

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□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□
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Prerequisite foundation

Fundamentals of Engineering Mathematics (1002) and Basics of Electrical and Electronics Engineering (1031).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
4	0	0	0	6 h/week; 90 h total	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Use power, energy, billing and conservation ideas for domestic applications.
2. Explain conversion of electrical energy to light, heat, chemical and mechanical forms.
3. Apply magnetism and magnetic-circuit principles.
4. Understand BJT operation, characteristics, configuration and biasing.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO ▯▯▯▯▯▯ exam-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. Define, explain, apply ▯▯▯▯▯ action words ▯▯▯▯▯▯▯▯▯▯▯▯▯.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Apply energy-efficiency criteria to choose household appliances for conservation.	Apply
CO2	Explain devices converting electrical energy to light, heat, chemical and mechanical forms.	Understand
CO3	Apply magnetism to magnetic materials, circuits and magnetization characteristics.	Apply
CO4	Compute BJT-bias resistance values for a specified operating point.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Electrical Energy Consumption and Conservation	Electrical power/energy, consumption, billing, conservation, BEE ratings and efficient appliance selection.	15 hours	CO1	Module 1
2	Applications of Electrical Energy	Electrical-to-light, heat, chemical and mechanical conversion: lamps, heating methods, electrolysis, solenoids, relays and DC motors.	15 hours	CO2	Module 2
3	Magnetism	Magnetic terms/materials, magnetic circuits, series/parallel circuits, air gaps, B–H curve and hysteresis.	15 hours	CO3	Module 3
4	Bipolar Junction Transistor	BJT modes, α/β, characteristics, CB/CE/CC configurations, biasing and stability.	15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Electrical Energy Consumption and Conservation

Electrical power/energy, consumption, billing, conservation, BEE ratings and efficient appliance selection.

CO1 · 15 hours

Applications of Electrical Energy

Electrical-to-light, heat, chemical and mechanical conversion: lamps, heating methods, electrolysis, solenoids, relays and DC motors.

CO2 · 15 hours

Magnetism

Magnetic terms/materials, magnetic circuits, series/parallel circuits, air gaps, B–H curve and hysteresis.

CO3 · 15 hours

Bipolar Junction Transistor

BJT modes, α/β, characteristics, CB/CE/CC configurations, biasing and stability.

CO4 · 15 hours

MODULE 1

Electrical Energy Consumption and Conservation

Electrical power/energy, consumption, billing, conservation, BEE ratings and efficient appliance selection.

15 hours | CO1 | Understand · Apply

Learning goals

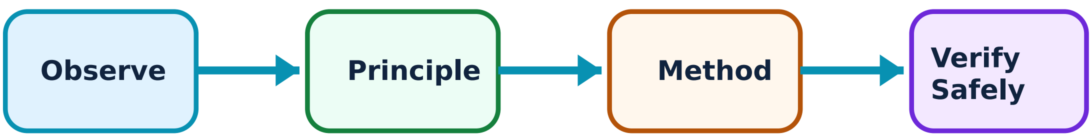
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electrical Energy Consumption and Conservation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Power, energy and consumption−

Electrical power is rate of energy conversion and electrical energy is accumulated use over time. Domestic consumption is obtained by combining appliance power rating with operating duration, then using consistent units.

Study and application method

List appliance rating, quantity and hours; calculate Wh or kWh, then sum daily/monthly energy before interpreting result.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List appliance rating, quantity and hours; calculate Wh or kWh, then sum daily/monthly energy before interpreting result.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse W with kWh; watt is power while kWh is energy.

2. Electricity billing and conservation−

A domestic bill reflects metered energy under tariff structure. Conservation reduces unnecessary use through correct operation, efficient appliance choice and avoiding standby loss without compromising required service.

Study and application method

Read units, tariff slabs/charges as provided and show bill calculation step by step. Propose measurable consumption-reduction actions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

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Common mistake / precaution: Use the applicable tariff supplied in a problem; do not assume a universal rate.

3. BEE star rating and appliance selection−

Star labels support comparison of appliance energy performance. Selection should consider service need, operating hours, energy use and life-cycle cost in the stated context.

Study and application method

Compare like-for-like appliances using rating and relevant capacity/usage, then state reasoned selection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare like-for-like appliances using rating and relevant capacity/usage, then state reasoned selection.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A star rating is meaningful only when comparing appropriate appliance category and capacity.

Energy-use calculator

Appliance power (W)

100

Hours used

5

Tariff per unit

6

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Applications of Electrical Energy

Electrical-to-light, heat, chemical and mechanical conversion: lamps, heating methods, electrolysis, solenoids, relays and DC motors.

15 hours | CO2 | Understand · Apply

Learning goals

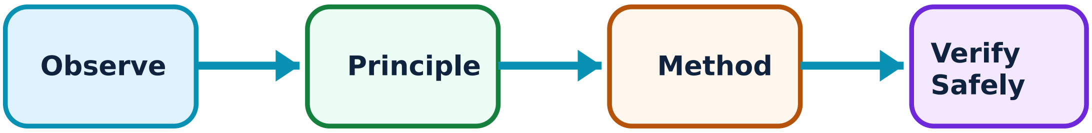
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Applications of Electrical Energy: identify the system, state its principle, follow the correct method and verify the result safely.

1. Lighting: incandescent, fluorescent and LED lamps−

Electrical energy can produce light by filament heating, gas-discharge/fluorescent processes or semiconductor electroluminescence. Construction and working determine efficiency, life and application.

Study and application method

Use a labelled construction sketch, state energy-conversion path, working sequence and one advantage/limitation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a labelled construction sketch, state energy-conversion path, working sequence and one advantage/limitation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not compare lamp efficiency only by brightness; rating, luminous output and usage condition matter.

2. Electric heating methods−

Resistance heating converts electrical energy to heat in an element. Arc, induction and dielectric heating use different physical mechanisms and suit different materials/processes.

Study and application method

For each method state principle, basic arrangement, one application and relevant safety or control point.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For each method state principle, basic arrangement, one application and relevant safety or control point.

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Common mistake / precaution: Heating methods are selected by material and process; not every method suits every load.

3. Electrolysis and electromechanical devices−

Electrolysis uses electrical energy for chemical change; Faraday’s laws relate chemical change to charge. Solenoids and relays use electromagnetic action, while a DC motor converts electrical energy to mechanical rotation.

Study and application method

Draw current path and moving/chemical part. Explain applications such as plating, refining, battery charging, switching or drive.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw current path and moving/chemical part. Explain applications such as plating, refining, battery charging, switching or drive.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrolytic and magnetic devices require correct polarity, ratings and isolation.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Magnetism

Magnetic terms/materials, magnetic circuits, series/parallel circuits, air gaps, B–H curve and hysteresis.

15 hours | CO3 | Understand · Apply

Learning goals

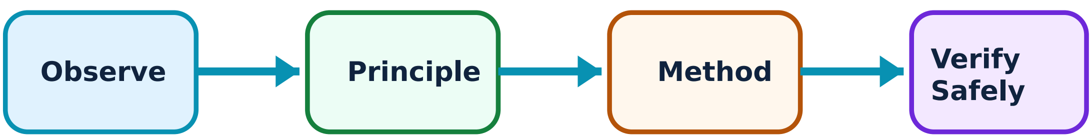
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Magnetism: identify the system, state its principle, follow the correct method and verify the result safely.

1. Magnetic terms and materials−

Magnetic field, flux, flux density, field strength and permeability describe magnetic behaviour. Diamagnetic, paramagnetic and ferromagnetic materials respond differently to applied field.

Study and application method

State symbol/unit where required, distinguish material groups with an example and relate permeability to magnetic-circuit design.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State symbol/unit where required, distinguish material groups with an example and relate permeability to magnetic-circuit design.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Flux density and field strength are different quantities; do not interchange their units.

2. Magnetic circuits and analogies−

A magnetic circuit provides a path for flux. Magnetomotive force drives flux against reluctance. Series and parallel magnetic paths can be analysed by a controlled analogy to electrical circuits.

Study and application method

Draw flux path, list length/area/material/air gap, calculate reluctance path by path and apply flux/ mmf relation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw flux path, list length/area/material/air gap, calculate reluctance path by path and apply flux/ mmf relation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrical analogy is useful but not exact; saturation changes magnetic behaviour.

3. Magnetization and hysteresis−

The B–H curve describes magnetization. A hysteresis loop shows lagging response and associated energy loss, important in electrical-machine core-material choice.

Study and application method

Sketch labelled axes/loop, state retentivity/coercivity and connect loop area qualitatively to hysteresis loss.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Method: Sketch labelled axes/loop, state retentivity/coercivity and connect loop area qualitatively to hysteresis loss.

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Common mistake / precaution: Do not treat hysteresis loss as the only magnetic loss in all machines.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Bipolar Junction Transistor

BJT modes, α/β, characteristics, CB/CE/CC configurations, biasing and stability.

15 hours | CO4 | Understand · Apply

Learning goals

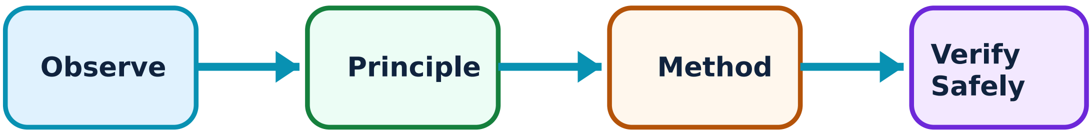
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bipolar Junction Transistor: identify the system, state its principle, follow the correct method and verify the result safely.

1. BJT modes, gains and characteristics−

Cutoff, active and saturation modes correspond to different junction-bias conditions. Current gains α and β relate transistor currents. Input/output/transfer characteristics show current-voltage behaviour.

Study and application method

Draw terminal-labelled symbol and characteristic axes, state operating mode and use correct α–β relationship for simple numerical work.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw terminal-labelled symbol and characteristic axes, state operating mode and use correct α–β relationship for simple numerical work.

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Common mistake / precaution: Do not use an amplifier active-region model when transistor is intentionally saturated as a switch.

2. CB, CE and CC configurations−

Common-base, common-emitter and common-collector circuits use different terminal as common reference and differ in current/voltage gain, input/output resistance and use.

Study and application method

Draw each configuration, identify input/output terminals and compare properties in one table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw each configuration, identify input/output terminals and compare properties in one table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Configuration name is determined by common terminal, not by device orientation in a drawing.

3. Biasing and stability−

Biasing establishes Q point for predictable transistor operation. Fixed, collector-to-base and voltage-divider bias circuits are used to compute resistance values for a given operating point; stability refers to resistance of Q point to variations.

Study and application method

Write desired Q point, supply and gain data, select stated bias method and solve resistor values with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

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Common mistake / precaution: Temperature and leakage affect bias; verify assumptions rather than treating gain as perfectly constant.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Electrical Energy Consumption and Conservation

Use the concept flow to revise observation, principle, method and verification.

Module 2: Applications of Electrical Energy

Use the concept flow to revise observation, principle, method and verification.

Module 3: Magnetism

Use the concept flow to revise observation, principle, method and verification.

Module 4: Bipolar Junction Transistor

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Calculate power/energy/bills and compare energy-efficiency practices.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare lamps, heating methods, electrolysis and relay/motor applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Solve series/parallel magnetic-circuit problems and interpret B–H/hysteresis curves.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Study BJT parameter effects, Q-point shift and bias stability.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Electrical Energy Consumption and Conservation

Explain Power, energy and consumption and state one correct method, application or precaution.−

Electrical power is rate of energy conversion and electrical energy is accumulated use over time. Domestic consumption is obtained by combining appliance power rating with operating duration, then using consistent units.

Answer framework: List appliance rating, quantity and hours; calculate Wh or kWh, then sum daily/monthly energy before interpreting result.

Important point: Do not confuse W with kWh; watt is power while kWh is energy.

Explain Electricity billing and conservation and state one correct method, application or precaution.−

A domestic bill reflects metered energy under tariff structure. Conservation reduces unnecessary use through correct operation, efficient appliance choice and avoiding standby loss without compromising required service.

Answer framework: Read units, tariff slabs/charges as provided and show bill calculation step by step. Propose measurable consumption-reduction actions.

Important point: Use the applicable tariff supplied in a problem; do not assume a universal rate.

Explain BEE star rating and appliance selection and state one correct method, application or precaution.−

Star labels support comparison of appliance energy performance. Selection should consider service need, operating hours, energy use and life-cycle cost in the stated context.

Answer framework: Compare like-for-like appliances using rating and relevant capacity/usage, then state reasoned selection.

Important point: A star rating is meaningful only when comparing appropriate appliance category and capacity.

Module 2 — Applications of Electrical Energy

Explain Lighting: incandescent, fluorescent and LED lamps and state one correct method, application or precaution.−

Electrical energy can produce light by filament heating, gas-discharge/fluorescent processes or semiconductor electroluminescence. Construction and working determine efficiency, life and application.

Answer framework: Use a labelled construction sketch, state energy-conversion path, working sequence and one advantage/limitation.

Important point: Do not compare lamp efficiency only by brightness; rating, luminous output and usage condition matter.

Explain Electric heating methods and state one correct method, application or precaution.−

Resistance heating converts electrical energy to heat in an element. Arc, induction and dielectric heating use different physical mechanisms and suit different materials/processes.

Answer framework: For each method state principle, basic arrangement, one application and relevant safety or control point.

Important point: Heating methods are selected by material and process; not every method suits every load.

Explain Electrolysis and electromechanical devices and state one correct method, application or precaution.−

Electrolysis uses electrical energy for chemical change; Faraday’s laws relate chemical change to charge. Solenoids and relays use electromagnetic action, while a DC motor converts electrical energy to mechanical rotation.

Answer framework: Draw current path and moving/chemical part. Explain applications such as plating, refining, battery charging, switching or drive.

Important point: Electrolytic and magnetic devices require correct polarity, ratings and isolation.

Module 3 — Magnetism

Explain Magnetic terms and materials and state one correct method, application or precaution.−

Magnetic field, flux, flux density, field strength and permeability describe magnetic behaviour. Diamagnetic, paramagnetic and ferromagnetic materials respond differently to applied field.

Answer framework: State symbol/unit where required, distinguish material groups with an example and relate permeability to magnetic-circuit design.

Important point: Flux density and field strength are different quantities; do not interchange their units.

Explain Magnetic circuits and analogies and state one correct method, application or precaution.−

A magnetic circuit provides a path for flux. Magnetomotive force drives flux against reluctance. Series and parallel magnetic paths can be analysed by a controlled analogy to electrical circuits.

Answer framework: Draw flux path, list length/area/material/air gap, calculate reluctance path by path and apply flux/ mmf relation.

Important point: Electrical analogy is useful but not exact; saturation changes magnetic behaviour.

Explain Magnetization and hysteresis and state one correct method, application or precaution.−

The B–H curve describes magnetization. A hysteresis loop shows lagging response and associated energy loss, important in electrical-machine core-material choice.

Answer framework: Sketch labelled axes/loop, state retentivity/coercivity and connect loop area qualitatively to hysteresis loss.

Important point: Do not treat hysteresis loss as the only magnetic loss in all machines.

Module 4 — Bipolar Junction Transistor

Explain BJT modes, gains and characteristics and state one correct method, application or precaution.−

Cutoff, active and saturation modes correspond to different junction-bias conditions. Current gains α and β relate transistor currents. Input/output/transfer characteristics show current-voltage behaviour.

Answer framework: Draw terminal-labelled symbol and characteristic axes, state operating mode and use correct α–β relationship for simple numerical work.

Important point: Do not use an amplifier active-region model when transistor is intentionally saturated as a switch.

Explain CB, CE and CC configurations and state one correct method, application or precaution.−

Common-base, common-emitter and common-collector circuits use different terminal as common reference and differ in current/voltage gain, input/output resistance and use.

Answer framework: Draw each configuration, identify input/output terminals and compare properties in one table.

Important point: Configuration name is determined by common terminal, not by device orientation in a drawing.

Explain Biasing and stability and state one correct method, application or precaution. −

Biasing establishes Q point for predictable transistor operation. Fixed, collector-to-base and voltage-divider bias circuits are used to compute resistance values for a given operating point; stability refers to resistance of Q point to variations.

Answer framework: Write desired Q point, supply and gain data, select stated bias method and solve resistor values with units.

Important point: Temperature and leakage affect bias; verify assumptions rather than treating gain as perfectly constant.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Electrical Energy Consumption and Conservation.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Applications of Electrical Energy.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Magnetism.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Bipolar Junction Transistor.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Electrical power/energy, consumption, billing, conservation, BEE ratings and efficient appliance selection..
2. Develop a complete answer or practical plan for: Electrical-to-light, heat, chemical and mechanical conversion: lamps, heating methods, electrolysis, solenoids, relays and DC motors..

3. Develop a complete answer or practical plan for: Magnetic terms/materials, magnetic circuits, series/parallel circuits, air gaps, B–H curve and hysteresis..
4. Develop a complete answer or practical plan for: BJT modes, α/β, characteristics, CB/CE/CC configurations, biasing and stability..

LAST-MILE REVISION

Quick Revision

Module 1: Electrical Energy Consumption and Conservation

Electrical power/energy, consumption, billing, conservation, BEE ratings and efficient appliance selection.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Applications of Electrical Energy

Electrical-to-light, heat, chemical and mechanical conversion: lamps, heating methods, electrolysis, solenoids, relays and DC motors.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Magnetism

Magnetic terms/materials, magnetic circuits, series/parallel circuits, air gaps, B–H curve and hysteresis.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Bipolar Junction Transistor

BJT modes, α/β, characteristics, CB/CE/CC configurations, biasing and stability.

- Match each topic to CO4.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Power, energy and consumption	Electrical power is rate of energy conversion and electrical energy is accumulated use over time.
Electricity billing and conservation	A domestic bill reflects metered energy under tariff structure.
BEE star rating and appliance selection	Star labels support comparison of appliance energy performance.
Lighting: incandescent, fluorescent and LED lamps	Electrical energy can produce light by filament heating, gas-discharge/fluorescent processes or semiconductor electroluminescence.
Electric heating methods	Resistance heating converts electrical energy to heat in an element.
Electrolysis and electromechanical devices	Electrolysis uses electrical energy for chemical change; Faraday’s laws relate chemical change to charge.
Magnetic terms and materials	Magnetic field, flux, flux density, field strength and permeability describe magnetic behaviour.
Magnetic circuits and analogies	A magnetic circuit provides a path for flux.
Magnetization and hysteresis	The B–H curve describes magnetization.
BJT modes, gains and characteristics	Cutoff, active and saturation modes correspond to different junction-bias conditions.
CB, CE and CC configurations	Common-base, common-emitter and common-collector circuits use different terminal as common reference and differ in current/voltage gain, input/output resistance and use.
Biasing and stability	Biasing establishes Q point for predictable transistor operation.

LEARNING RESOURCES

References

Officially listed print resources

1. B. L. Theraja and A. K. Theraja, Electrical Technology Vol. I.
2. V. K. Mehta and Rohit Mehta, Principles of Electrical Engineering.
3. Robert Boylestad and Louis Nashelsky, Electronic Devices and Circuit Theory.

Officially listed web resources

- https://beeindia.gov.in

- https://www.electrical4u.com
- https://www.electronics-tutorials.ws

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